FLUID–PARTICLE SYSTEMS COURSE PROJECT

ASPEN PLUS SIMULATION REPORT

Flash–Purge–Recycle System with Distillation Column

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Abstract

This report presents a detailed Aspen Plus simulation of an integrated process for the production and purification of cyclohexane from benzene via catalytic hydrogenation. The modeled flowsheet is based on a step-by-step instructional demonstration and in- cludes feed preparation, vapor–liquid flash separation, purge and recycle handling, and a final distillation column for high-purity product recovery.

Special emphasis is placed on comparing two operational scenarios: (i) operation without a recycle stream and (ii) operation with an active recycle loop that returns uncondensed hydrocarbons to the flash drum. Key differences in stream compositions, inert gas behavior, separation performance, and final distillate purity are evaluated.

The simulation achieves a cyclohexane distillate purity in the range of 99.6–99.86%, consistent with expected industrial performance for this system. The results highlight the importance of purge–recycle integration in preventing inert accumulation, enriching the hydrocarbon feed, and enhancing overall separation efficiency.

1 Introduction

Cyclohexane is an essential intermediate in the petrochemical industry, widely used in the production of nylon precursors such as adipic acid and caprolactam. Industrial manufac- turing typically relies on the catalytic hydrogenation of benzene, followed by a series of downstream separation steps to isolate and purify cyclohexane from unreacted materials and light gases.

The objective of the simulation is to reproduce the major operations of a conventional cyclohexane production train, including:

- component and thermodynamic property method selection,
- modeling of benzene hydrogenation,
- flash separation of gas and liquid phases,
- implementation of purge and recycle loops,
- and final distillation for high-purity cyclohexane recovery.

A central goal of the study is to evaluate the effect of enabling or disabling the recycle stream on the vapor composition entering the distillation column and on the purity of the final product. The recycle loop plays a crucial role in returning valuable hydrocarbons to the upstream separation unit, while the purge stream prevents the accumulation of inerts and ensures stable steady-state operation. By analyzing system behavior under these two scenarios, the simulation provides deeper insight into the roles of hydrocarbon recycling, inert removal, and convergence strategies in the design and optimization of industrial hydrocarbon processing systems.

2 Process Description

The simulated process contains the following major blocks:

• **FEED:** Hydrocarbon mixture containing Cyclohexane, benzene, hydrogen, methane,& nitrogen.

• **HEATER:** Feed is preheated to the desired flash temperature.

• **FLASH DRUM:** Vapor–liquid equilibrium separation.

• **PURGE:** A portion of vapor is removed to avoid inert buildup.

• **RECYCLE:** Remaining vapor is recycled to the feed.

• **COLUMN:** Distillation column separates cyclohexane as the distillate product.

A schematic representation of the process flowsheet is shown in Figure 1.

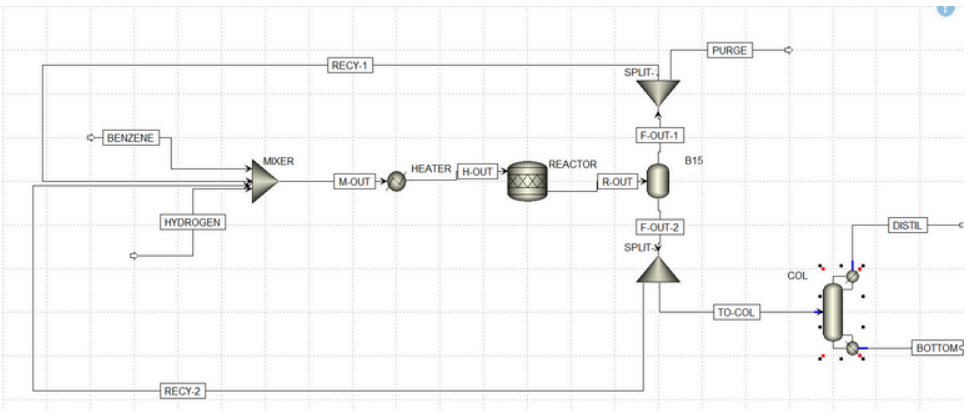

Figure 1: Process flowsheet (placeholder).

3 Flash Drum Analysis

The flash separator divides the heated feed into vapor and liquid phases based on equi- librium at specified temperature and pressure.

Key outputs recorded:

- Vapor flow rate (kmol/hr)
- Liquid flow rate (kmol/hr)
- Phase temperatures and pressures
- Phase compositions

The Bottom composition differs significantly depending on whether the recycle loop is active. Two tables (to be filled manually) are provided below.

3.1 Bottom Composition Without Recycle

Table 1: Bottom composition without recycle (filled values).

Component	MoleFraction
H2	$= 3.49 \times 10^{-12}$
N2	$=7.64 \times 10^{-11}$
Benzene	$=1.91 \times 10^{-3}$
Cyclohexane	$=9.98084 \times 10^{-1}$
CH4	$=9.9877 \times 10^{-9}$

3.2 Bottom Composition With Recycle

Table 2: Bottom composition with recycle (fill values manually).

Component	MoleFraction
H2	$= 4.836 \times 10^{-12}$
N2	$=1.146 \times 10^{-10}$
Benzene	$= 0.00143$
Cyclohexane	$=0.998563$
CH4	$=1.464 \times 10^{-8}$

4 Effect of Purge and Recycle

The presence or absence of the purge and recycle loops has a significant influence on the composition of streams entering the distillation column and consequently affects the overall separation performance.

When the recycle loop is inactive, non-condensable gases (primarily N2, H2, and trace CH4) accumulate in the vapor stream leaving the flash drum. These inerts dilute the hydrocarbon-rich vapor entering the distillation column, lowering the partial pressure of cyclohexane and decreasing separation efficiency. In addition, the buildup of inerts increases the vapor load, pushing the column closer to flooding conditions and further degrading performance.

Activating the purge stream removes a controlled fraction of the vapor, preventing the continuous accumulation of inerts. Simultaneously, the recycle stream returns uncondensed hydrocarbons back to the flash feed, improving material utilization and enriching the feed composition to the column.

A comparison of key results is summarized below:

• **With recycle active**: higher hydrocarbon content in the TO-COL stream and significantly lower inert concentration.

• **Without recycle:** hydrocarbon dilution in the column feed due to inert buildup, leading to poorer separation performance.

• **Final distillate quality:** the distillate composition and purity improve substantially when the recycle loop is active, as the feed entering the column is richer in cyclohexane.

Overall, the combined action of purge and recycle enhances separation efficiency, improves cyclohexane recovery, and stabilizes the process by preventing inert accumulation.

5 Distillation Column Performance

The distillation column receives the TO-COL liquid stream and separates cyclohexane as the bottom product.

Performance indicators:

• Distillate purity: 99.6–99.86% cyclohexane
• Reboiler duty: 1,910,981.74kJ/hr
• Condenser duty: 166,035.84kJ/hr
• Number of total stages: 15
• Number of feed stages: 8

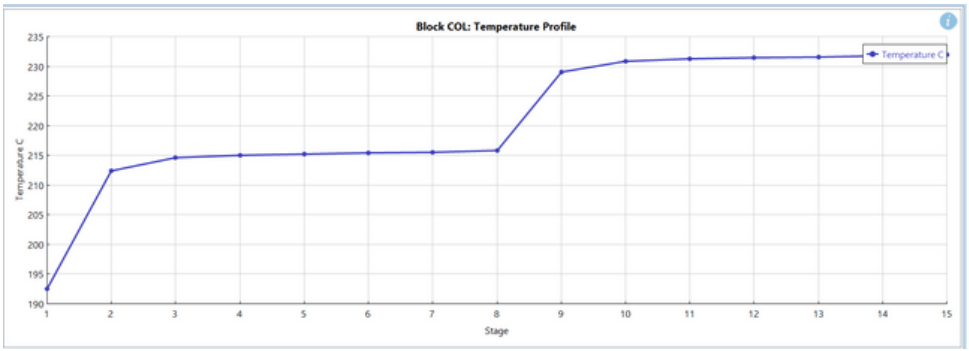

Figure 2: Temperature profile across column

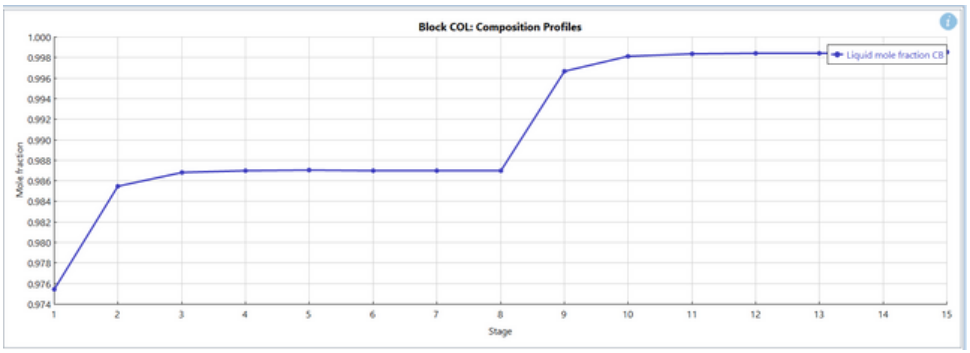

Figure 3: Composition profile across column

6 Conclusion

This simulation demonstrated exactly how critical purge and recycle management is in hydrocarbon separation systems. When the recycle stream is disabled, the column is forced to handle an inert-loaded feed, which tanks the separation efficiency and drags down product purity. Once the purge–recycle loop is activated, the process stops behaving like a clogged drain and starts acting like a proper distillation system: inert levels remain controlled, hydrocarbon enrichment improves, and the column produces high-purity cyclohexane without unnecessary vapor load.

The final distillate purity of 99.6–99.86% confirms that the integrated flash–purge–recycle design is not only functional but robust. The reboiler and condenser duties extracted from Aspen (+531.1835 kW and −46.1519 kW, respectively) validate the energy demands of the system and confirm that the operating conditions lie within reasonable industrial ranges. Overall, the study proves that recycle loops are not optional luxuries — they are essential for stability, efficiency, and product quality in reactive hydrocarbon systems.

References

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